

SYSTEM DESIGN DOCUMENT: ALIGNED SURVEYORS WEB APP

1. System Overview

The Aligned Surveyors Portal is a split-domain web application built to serve two core purposes:

1. **Public Site:** A sleek showcase of the company's fields of practice (Topographical, Mine, Engineering surveys, etc.) and project portfolio (e.g., Mimosa TSF4, Mazowe Mall, IDBZ projects), embedded with an autonomous, data-aware Gemini Chat Assistant to handle visitor inquiries.
2. **Internal Workspace:** A private, secure administrative dashboard providing real-time team communication (Laravel Reverb) and text-only project tracking updates for staff.

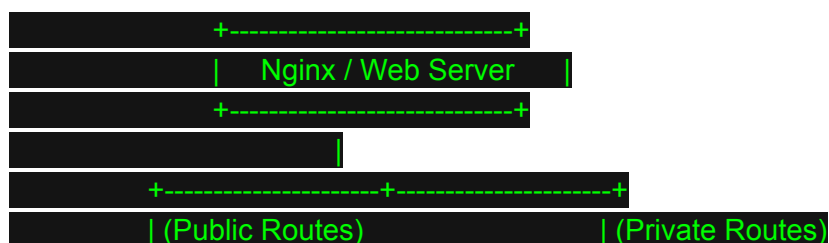
The system utilizes strict domain isolation, a bare-metal deployment structure (no Docker in production), and leverages custom secure API-driven infrastructure for communication to bypass VPS hosting limitations.

2. Technology Stack & Environment

- **Local Development Environment:** Ubuntu Linux
- **Production Deployment Environment:** Bare-Metal Linux Virtual Private Server (VPS) via Render / Direct Deployment
- **Backend Framework:** Laravel 12
- **Frontend Architecture:** React + Inertia.js
- **Database Engine:** Supabase (PostgreSQL)
- **Real-time Engine:** Laravel Reverb (WebSocket Server)
- **AI Engine:** Google Gemini API
- **Mail Subsystem:** Google OAuth API Transport (Mail::extend)

3. Architecture & Domain Isolation Matrix

To keep the codebase modular, routing and business logic are divided into isolated layers within app/Domains/. Cross-domain leakage is strictly prohibited.





3.1. Domain/Marketing

- **Routes File:** routes/tenant/marketing.php
- **Responsibilities:**
 - Serving the landing page, operational capabilities, asset registry (e.g., Topcon Hiper II, DJI Phantom 4 Pro) , and historic proof of works.
 - Handling public intake requests for booking consultations.
 - Exposing a secure endpoint for the client-facing Gemini widget.

3.2. Domain/Auth

- **Routes File:** routes/tenant/auth.php (Customized from Breeze)
- **Responsibilities:**
 - Enforcing access control via a hidden, unlinked portal entry path.
 - Restricting account registration to authenticated administrators.
 - Intercepting default password reset/verification notifications to process through standard HTTPS API requests instead of traditional SMTP mail protocols.

3.3. Domain/Internal

- **Routes File:** routes/tenant/internal.php
- **Responsibilities:**
 - Providing internal project boards where administrators and engineers can append chronological progress updates.

- Powering real-time peer-to-peer and group chat interfaces via WebSockets.

4. System Components & Integration Architecture

4.1. The AI Integration Blueprint (Gemini Chatbot)

The client-facing assistant must provide accurate responses regarding company history, capabilities, and previous contracts.

- **Context Ingestion Pipeline:** The chatbot controller does not rely on static document uploads or external RAG software. It queries local tables (services, projects, assets) using predefined read-only Eloquent scopes.
- **System Prompt Constraints:**

Plaintext

You are the Aligned Surveyors Virtual Assistant. Your knowledge is strictly limited to the data provided below.

If asked about a survey type, client, or equipment asset not listed in the context matrix, state clearly that you do not have record of it and prompt them to book a human consultation.

4.2. Secure Email Pipeline (Google OAuth API Transport)

Standard cloud VPS providers routinely block ports 25, 465, and 587 to prevent spam vector vulnerabilities. To ensure Breeze features function perfectly without requiring standard relays, the mail backend is custom-extended.

- **Mechanism:** Using a valid Google Developer Client ID, Secret, and an offline Refresh Token, the platform updates an Access Token in real-time.
- **Payload Delivery:** Outbound transactional emails are transformed into standard RFC 2822 formatting, converted to a web-safe base64Url payload, and dispatched securely to <https://gmail.googleapis.com/gmail/v1/users/me/messages/send> over HTTPS Port 443.

4.3. Real-Time Layer (Laravel Reverb)

- **Engine:** Reverb runs natively as a persistent daemon process on the VPS.
- **Channels:**
 - `presence-chat.room.{id}`: Manages localized internal chat threads (Direct & Team Group Messaging).
 - `private-project.tracker.{id}`: Dispatches shallow text notifications when a project milestone or log update is captured.

5. Security & Access Control Policy

- **Obscurity Strategy:** There is no "Login" button or anchor text anywhere on the public-facing site. Access is guarded by a discreet URL path.
- **Initialization Safeguard:** The application seed array contains exactly one Root Administrator entry, drawing its structural values (INITIAL_ADMIN_EMAIL, INITIAL_ADMIN_PASSWORD) straight from the host machine's environment configurations.
- **User Lifecycle Control:** * Only an active Admin possesses the authorization rights to trigger RegisteredUserController::store().
 - New team accounts do not generate automated session logins upon backend creation.
 - Admins can instantly terminate active sessions and lock individual components by toggling a user row's is_active status to false.

6. Functional Execution Roadmaps (Single-Loop Iteration Model)

In alignment with your exact execution rule, features are built sequentially inside an isolated **ONE Model -> ONE Controller -> ONE View** loop. No subsequent loop may begin until the active cycle satisfies testing parameters.

1. **Loop 1: Custom Google Mail Transport Layer** (*Completed*)
2. **Loop 2: Admin Seeding & Locked-Down Registration** (*Completed*)
3. **Loop 3: Services Capability Architecture** (Managing fields of practice mapping to the corporate profile data)
4. **Loop 4: Past Projects Repository & Showcase View** (Setting up historical contract logging for chatbot access)
5. **Loop 5: Gemini Chat Engine API Integration**
6. **Loop 6: Consultation Ingestion Pipeline**
7. **Loop 7: Internal Project Logs Engine** (Text-only system updates)
8. **Loop 8: Reverb Real-Time Group & Peer Communication Subsystem**